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FACULTY OF INFORMATION ENGINEERING,
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DEPARTMENT OF COMPUTER, CONTROL
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MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE AND ROBOTICS

Reinforcement Learning

Lecture notes

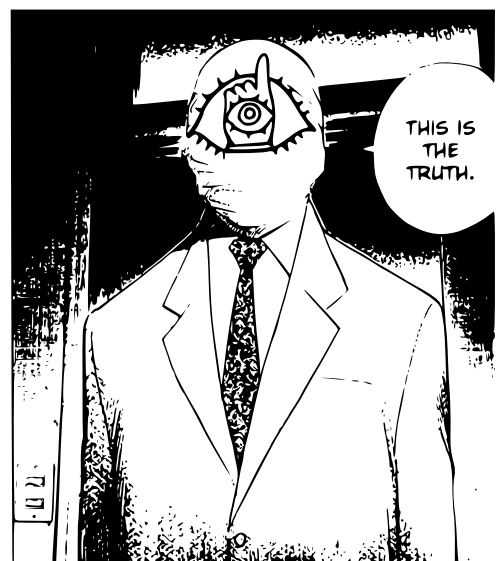
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This document summarizes and presents the topics for the Reinforcement Learning course for the Master's degree in Artificial Intelligence and Robotics at Sapienza University of Rome. The document is free for any use. If the reader notices any typos, they are kindly requested to report them to the author.



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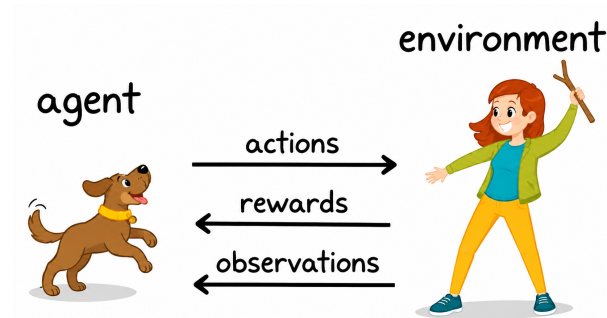
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INTRODUCTION

Definition 1.0.1 : Reinforcement Learning

With **Reinforcement Learning** we mean the process to *learn* what to do in order to maximize a numerical reward signal:

- The learner must discover which actions yield the highest reward by trying them.
- The actions might affect not only immediate reward, but also subsequent ones. It is a *trial-and-error* process.



The agent has to:

- **Exploit** what it has already experienced to obtain a reward.
- **Explore** in order to make better action selections in the future.

One of the main challenges of reinforcement learning is the exploration-exploitation trade-off. In stochastic tasks, each action must be tried many times, in order to correctly estimate the expected reward.

1.1 Markov Decision Processes

First of all we introduce a *sequential* decision making approach. We see the agent that has to learn the policy as an entity that can interact with the world/environment through actions a_t , and can receive

from the environment an observation o_t and a reward r_t after each action. The agent interacts at discrete time steps; such discrete interaction generates a **trajectory** (or history) of tuples (observation-action-reward):

$$\underbrace{h_t}_{\text{history at time } t} = (o_0, a_0, r_1, o_1, a_1, r_2, \dots, r_t, o_t, a_t). \quad (1.1)$$

We define a state function $s_t = f(h_t)$ that describes the current state of the agent at time t and depends on the history h_t and is typically hidden or unknown.

Definition 1.1.1 : Markov Assumption

A state s_t is **Markovian** if the future is independent of the past information, given the present state: the probability of ending up in a given state s_{t+1} by applying a_t at time t , depends only on the current state s_t and not on the history h_t :

$$\mathbb{P}(s_{t+1} \mid s_t, a_t) = \mathbb{P}(s_{t+1} \mid h_t, a_t). \quad (1.2)$$

A state can always be made Markovian if we define the state exactly as the previous history:

$$s_t = h_t. \quad (1.3)$$

In practice the current state is defined as the *latest observation*:

$$s_t = o_t \quad (1.4)$$

in this case the state is said to be **fully observable**.

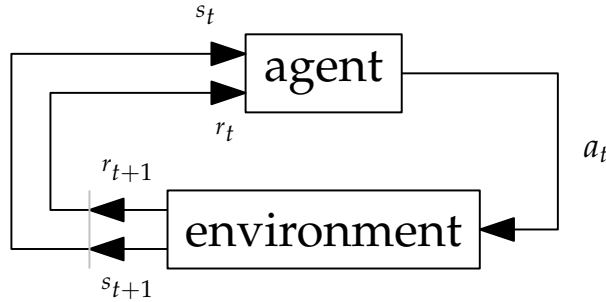


Figure 1.1: Markov Decision Process scheme.

A **Markov Decision Process (MDP)** is a sequential decision making scheme under the Markov assumption. The transition dynamics is generally stochastic: $\mathbb{P}(s_{t+1} \mid s_t, a_t)$. An alternative notation to denote the fact that the next state is stochastic by applying a_t on s_t is the following:

$$s_{t+1} \sim \mathbb{P}(\cdot \mid s_t, a_t) \quad (1.5)$$

$$\text{or } s' \sim \mathbb{P}(\cdot \mid s, a) \quad (1.6)$$

A **reward** r_t is a scalar signal representing feedback on the actions taken by the agent, and indicates how well an agent is doing at a given time step t . The reward is a function of the state and action (reward taken by applying a given action at a given state):

$$r_{t+1} = R(s_t, a_t). \quad (1.7)$$

We define the **cost** as the opposite of the reward. Although in reinforcement learning the objective is always to maximize the reward function, the fact that this is the best way to reach a goal is not guaranteed, in fact the so-called **Reward Hypothesis**:

“all goals can be described as the maximization of expected cumulative reward.”

remains an open question today.

Definition 1.1.2 : Markov Decision Process

An MDP is defined by a 5-tuple:

$$(S, A, \mathbb{P}, R, \gamma) \quad (1.8)$$

where:

- S is the set of all possible states.
- A is the set of all the actions the agent can take.
- $\mathbb{P}$ is the transition probability, which defines the dynamics of the environment.
- R is the reward function.
- γ is the **discount factor**, a real value within the interval $[0, 1)$ that determines the importance of future rewards compared to immediate ones.

The behavior of an agent is given by a policy, a function that determines how agents select actions.

Definition 1.1.3 : Policy

A **policy** is a function $\pi : S \rightarrow \Delta(A)$ that assigns a distribution over the set of actions to every state.

A policy can be deterministic (special case where $\pi : S \rightarrow A$) and we write:

$$\pi(s) = a \quad (1.9)$$

or stochastic:

$$\pi(a | s) \in [0, 1] \quad \text{with} \quad \sum_{a \in A} \pi(a | s) = 1. \quad (1.10)$$

We can also write $a \sim \pi(\cdot | s)$. Let's define the **probability of a trajectory**: What's the probability of seeing a given trajectory at time t according to π starting at s_0 ? We have the trajectory

$$(s_0, a_0, s_1, \dots, s_t, a_t) \quad (1.11)$$

and

$$\mathbb{P}^\pi(a_0, s_1, \dots, s_t, a_t | s_0) \quad (1.12)$$

$$= \pi(a_0 | s_0) \mathbb{P}(s_1 | s_0, a_0) \pi(a_1 | s_1) \mathbb{P}(s_2 | s_1, a_1) \dots \mathbb{P}(s_t | s_{t-1}, a_{t-1}) \pi(a_t | s_t) \quad (1.13)$$

notice how we added π as superscript to denote the fact that this probability depends on the given policy function. We can also define the **state visitation probability**, i.e. the probability of visiting a particular state $\hat{s}$ at time t according to π starting from s_0 :

$$\mathbb{P}_t^\pi(s_t = \hat{s} | s_0) = \sum_{a_0, s_1, \dots, a_{t-1}} \mathbb{P}^\pi(s_0, a_0, s_1, \dots, s_t = \hat{s}). \quad (1.14)$$

The summation in equation (1.14) sums over all trajectories of length t that start at s_0 and end in $\hat{s}$.

A basic example of a Markov decision process is a robotic manipulator with a hand gripper that can grab and carry a ball:

- the states are the possible robot configurations (joint values) and the ball position,
- the actions are the possible torques on the arm and the finger joints,
- the transition is stochastic, given by the physics (how the torques update the joint values over time) and by the noise,
- the policy maps each configuration+ball position to a torque for the joints,
- the cost (opposite of the reward) is the magnitude of the torque and the distance to the goal.

1.1.1 MDP Horizon

In Definition 1.1.2 we introduced a discount factor $\gamma \in [0, 1]$. This term weights the importance of immediate rewards with respect to future rewards:

- $\gamma = 0$: Our agent only cares about immediate rewards.
- $\gamma = 1$: Immediate and future rewards are equally important.

The discount factor is used in the **Value Function** V^π , a function that estimates the goodness of a state based on its value. It is also a measure to compare different policies.

Definition 1.1.4 : Value Function

The **Value Function** V^π measures the cumulative reward (the sum of all future rewards, possibly discounted by the γ factor) starting from s and continuing to follow the policy π for all subsequent actions.

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{h=0}^{\infty} \gamma^h r_{h+1} \mid s_0 = s \right] \quad (1.15)$$

$$= \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h \cdot \underbrace{r_{h+1}}_{R(s_h, a_h)} \mid s_0 = s, a_h \sim \pi(\cdot \mid s_h), s_{h+1} \sim \mathbb{P}(\cdot \mid s_h, a_h) \right]. \quad (1.16)$$

With $\mathbb{E}$ we denote the expected value. The value function V^π considers all the possible infinite trajectories that are generated by applying π starting from the argument s , since the transition and the policy are (or might be) stochastic, a probability is assigned to each trajectory, so the expected value is considered, i.e. the sum over the cumulative reward of a given trajectory, weighted by the probability of that trajectory.

Example: Let's assume that there is an MDP with three possible trajectories starting from s_a :

$$\text{trajectory 1 : } (s_a, a, s_c) \quad (1.17)$$

$$\text{trajectory 2 : } (s_a, a, s_b, a, s_c) \quad (1.18)$$

$$\text{trajectory 3 : } (s_a, a, s_b, a, s_b, a, s_c) \quad (1.19)$$

There is only one action (always chosen by the policy) and the reward function is deterministic:

$$R(s_a, a) = 2 \quad (1.20)$$

$$R(s_b, a) = 1 \quad (1.21)$$

$$R(s_c, a) = 0 \quad (1.22)$$

Usually these trajectories are infinite, but in this case we assume that they all end up in s_c , an ending state where all actions have a reward of zero with probability 1:

$$\mathbb{P}(s_c \mid s_c, a) = 1. \quad (1.23)$$

The expected value of the cumulative reward assuming the following probabilities:

$$\mathbb{P}(s_a, a, s_c) = 0.2 \quad (1.24)$$

$$\mathbb{P}(s_a, a, s_b, a, s_c) = 0.1 \quad (1.25)$$

$$\mathbb{P}(s_a, a, s_b, a, s_b, a, s_c) = 0.7 \quad (1.26)$$

assuming $\gamma = 1$ is

$$2 \cdot 0.2 + 3 \cdot 0.1 + 4 \cdot 0.7 = 3.5. \quad (1.27)$$

The Action-Value function Q^π goes one step further than V^π , tying the value not only to the state, but also to the action you choose to take.

Definition 1.1.5 : Action-Value Function

The **Action-Value Function** Q^π measures the expected value of the cumulative reward starting from a state s , performing a specific action a and then continuing to follow the policy π .

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h \cdot \underbrace{r_{h+1}}_{R(s_h, a_h)} \mid (s_0, a_0) = (s, a), \underbrace{a_h \sim \pi(\cdot \mid s_h)}_{\forall h \geq 1}, s_{h+1} \sim \mathbb{P}(\cdot \mid s_h, a_h) \right]. \quad (1.28)$$

Theorem 1.1.1 : The Bellman Equation

The value of a certain state can be expanded in terms of the current reward and the value of the next states according to the policy:

$$V^\pi(s) = \sum_{a \in A} \pi(a \mid s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' \mid s, a) V^\pi(s') \right]. \quad (1.29)$$

The Bellman equation can be applied to the action-value function:

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' \mid s, a) V^\pi(s'). \quad (1.30)$$

In Equation (1.29) V^π is a function of s and $\pi(s)$ while in equation (1.30) Q^π is a function of s and a (as highlighted by the green characters). As a result of these equations we have the following identity:

$$V^\pi(s) = \sum_{a \in A} \pi(a \mid s) Q^\pi(s, a) \quad (1.31)$$

if π is deterministic:

$$V^\pi(s) = Q^\pi(s, \pi(s)) \quad (1.32)$$

V^π can be expressed in terms of Q^π . One last quantity that we introduce is the following.

Definition 1.1.6 : Discounted State-Action Distribution

The discounted state-action distribution quantifies how often (and with what discounted time weight) the pair (s, a) is visited along the entire infinite horizon.

$$d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}^\pi(s_h = s \mid s_0) \pi(a \mid s). \quad (1.33)$$

Since there is a discount factor $\gamma < 1$, events that occur far away in time matter less and less.

$\mathbb{P}^\pi(s_h = s \mid s_0) \pi(a \mid s)$ is the exact probability of being in state s and executing action a exactly at time step h , conditioned on having started in s_0 and having followed the π policy for all previous steps.

1.2 Optimality of a Policy

For an infinite horizon MDP there always exists a deterministic **optimal policy** π^* that satisfies:

$$V^{\pi^*}(s) \geq V^\pi(s), \quad \forall s, \forall \pi \quad (1.34)$$

This means that the policy π^* is the one that guarantees a greater reward (in expected value), since it dominates all other policies in each state. We will use also the following notation:

$$V^{\pi^*} = V^* \quad (1.35)$$

$$Q^{\pi^*} = Q^*. \quad (1.36)$$

The **Bellman Optimality** condition states that:

$$V^*(s) = \max_{a \in A} Q^*(s, a). \quad (1.37)$$

Example: Consider the MDP in Figure 1.2.

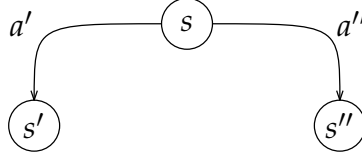


Figure 1.2: Three-states MDP example.

We assume that V^* is known in the states s' and s'' . We want to try a' and a'' in s in order to get the rewards and compute Q^* in s for the actions:

- We apply a' and get $R(s, a')$. We can compute $Q^*(s, a') = R(s, a') + \underbrace{\gamma V^*(s')}_{\text{known}}$.
- We apply a'' and get $R(s, a'')$. We can compute $Q^*(s, a'') = R(s, a'') + \underbrace{\gamma V^*(s'')}_{\text{known}}$.

Now we can use the identity in Equation (1.37) to compute $V^*(s)$:

$$V^*(s) = \max\{Q^*(s, a'), Q^*(s, a'')\}. \quad (1.38)$$

Theorem 1.2.1 : Bellman Optimality (Existence)

Let $\gamma \in [0, 1)$. The optimal value function $V^*(s)$ satisfies the nonlinear Bellman equation:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (1.39)$$

hence the policy:

$$\hat{\pi}(s) = \arg \max_{a \in A} Q^*(s, a) \quad (1.40)$$

is the optimal policy: $V^{\hat{\pi}} = V^*$.

Proof: First we prove that V^* satisfies the Bellman Equation (1.39). For a generic policy π , we can write the Bellman equation by isolating the first action $a \sim \pi(\cdot | s)$:

$$V^\pi(s) = \sum_{a \in A} \pi(a | s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^\pi(s') \right]. \quad (1.41)$$

The optimal value function can be written as follows:

$$V^*(s) = \max_{\pi} \sum_{a \in A} \pi(a | s) \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^\pi(s') \right]. \quad (1.42)$$

Consequently, the maximum over π coincides with the maximization over a single action $a \in A$ by placing $V^*(s')$ in the future steps:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (1.43)$$

hence V^* satisfies the Bellman equation. Now we have to prove that $\hat{\pi}$ is the optimal policy. We define the deterministic policy $\hat{\pi}$ as in Equation (1.40). Since $\hat{\pi}(s)$ selects the action that maximizes Q^* we can

write:

$$V^*(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right] \quad (1.44)$$

$$= R(s, \hat{\pi}(s)) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) V^*(s'). \quad (1.45)$$

On the other hand, from the Bellman expectation equation applied to $\hat{\pi}$ we can write:

$$V^{\hat{\pi}}(s) = R(s, \hat{\pi}(s)) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) V^{\hat{\pi}}(s') \quad (1.46)$$

if we subtract (1.46) from (1.45) we get:

$$V^*(s) - V^{\hat{\pi}}(s) = \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')). \quad (1.47)$$

Now we take the absolute value of both sides:

$$|V^*(s) - V^{\hat{\pi}}(s)| = \left| \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')) \right|. \quad (1.48)$$

By the triangle inequality:

$$|V^*(s) - V^{\hat{\pi}}(s)| = \left| \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) (V^*(s') - V^{\hat{\pi}}(s')) \right| \quad (1.49)$$

$$\leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) |V^*(s') - V^{\hat{\pi}}(s')| \quad (1.50)$$

$$\leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) \|V^* - V^{\hat{\pi}}\|_{\infty} \quad (1.51)$$

we used the L_{∞} norm, the following inequality holds:

$$|V^*(s) - V^{\hat{\pi}}(s)| \leq \gamma \sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) \|V^* - V^{\hat{\pi}}\|_{\infty}. \quad (1.52)$$

$\|V^* - V^{\hat{\pi}}\|_{\infty}$ is constant and $\sum_{s' \in S} \mathbb{P}(s' | s, \hat{\pi}(s)) = 1$, hence

$$|V^*(s) - V^{\hat{\pi}}(s)| \leq \gamma \|V^* - V^{\hat{\pi}}\|_{\infty}. \quad (1.53)$$

this holds for each s hence:

$$\|V^* - V^{\hat{\pi}}\|_{\infty} \leq \gamma \|V^* - V^{\hat{\pi}}\|_{\infty} \implies \quad (1.54)$$

$$\|V^* - V^{\hat{\pi}}\|_{\infty} = 0 \implies \quad (1.55)$$

$$V^{\hat{\pi}}(s) = V^*(s), \quad \forall s. \quad (1.56)$$

■

Theorem 1.2.2 : Bellman Optimality (Uniqueness)

Let $\gamma \in [0, 1)$. For any value function V , if

$$V(s) = \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right] \quad \forall s \in S \quad (1.57)$$

then $V(s) = V^*(s)$ for all $s \in S$.

Proof: Let V be any value function satisfying the Bellman optimality equation for all $s \in S$. Since V^* is also a solution, we can write (for any state s):

$$|V(s) - V^*(s)| \quad (1.58)$$

$$= \left| \max_{a \in A} \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right) - \max_{a \in A} \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right) \right|. \quad (1.59)$$

Using the property $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$:

$$|V(s) - V^*(s)| \quad (1.60)$$

$$\leq \max_{a \in A} \left| \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V(s') \right) - \left(R(s, a) + \gamma \sum_{s' \in S} \mathbb{P}(s' | s, a) V^*(s') \right) \right| \quad (1.61)$$

we can cancel out $R(s, a)$ and move γ outside the max function:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \left| \sum_{s' \in S} \mathbb{P}(s' | s, a) (V(s') - V^*(s')) \right| \quad (1.62)$$

by the triangle inequality we can move out the absolute value from the sum:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \sum_{s' \in S} \mathbb{P}(s' | s, a) |V(s') - V^*(s')|. \quad (1.63)$$

We use the definition of L_∞ norm:

$$|V(s') - V^*(s')| \leq \max_{y \in S} |V(y) - V^*(y)| = \|V - V^*\|_\infty \quad (1.64)$$

Equation (1.63) becomes:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \sum_{s' \in S} \mathbb{P}(s' | s, a) \|V - V^*\|_\infty \quad (1.65)$$

$\|V - V^*\|_\infty$ is a constant hence can be moved out of the summation:

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} \left(\|V - V^*\|_\infty \sum_{s' \in S} \mathbb{P}(s' | s, a) \right) \quad (1.66)$$

trivially

$$\sum_{s' \in S} \mathbb{P}(s' | s, a) = 1$$

hence

$$|V(s) - V^*(s)| \leq \gamma \max_{a \in A} (\|V - V^*\|_\infty \cdot 1) = \gamma \|V - V^*\|_\infty \quad (1.67)$$

since this inequality holds for each state $s \in S$, we can write:

$$\underbrace{\max_{s \in S} |V(s) - V^*(s)|}_{\|V - V^*\|_\infty} \leq \gamma \|V - V^*\|_\infty \implies \quad (1.68)$$

$$\|V - V^*\|_\infty \leq \gamma \|V - V^*\|_\infty \implies \quad (1.69)$$

$$(1 - \gamma) \|V - V^*\|_\infty \leq 0 \implies \|V - V^*\|_\infty = 0. \quad (1.70)$$

This holds since $\gamma \in [0, 1)$, hence $(1 - \gamma) > 0$. By definition:

$$\|V - V^*\|_\infty = 0 \implies \quad (1.71)$$

$$V(s) = V^*(s) \quad \forall s \in S. \quad (1.72)$$

■

This formulation reduces the multi-step problem to a one-step greedy choice plus the continuation value $V(s')$. By uniqueness of the fixed point, any function V satisfying this optimality equation coincides with V^* .